

RESEARCH ARTICLE | OCTOBER 05 2023

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Physics of Fluids 35, 103105 (2023)

<https://doi.org/10.1063/5.0166075>



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Cite as: Phys. Fluids **35**, 103105 (2023); doi: [10.1063/5.0166075](https://doi.org/10.1063/5.0166075)

Submitted: 3 July 2023 · Accepted: 16 September 2023 ·

Published Online: 5 October 2023



View Online



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M. Shan Tariq,¹ W. Masood,^{1,2} M. Siddiq,² S. Asghar,³ B. M. Alotaibi,⁴ Sherif M. E. Ismaeel,^{5,6} and S. A. El-Tantawy^{7,8,a)}

AFFILIATIONS

¹Department of Physics, COMSATS University Islamabad, Islamabad, Pakistan

²National Center for Physics (NCP), Quaid-i-Azam University Campus, Shahdrah Valley Road, Islamabad 44000, Pakistan

³Department of Mathematics, COMSATS University Islamabad, Islamabad, Pakistan

⁴Department of Physics, College of Science, Princess Nourah bint Abdulrahman University, P.O. Box 84428, Riyadh 11671, Saudi Arabia

⁵Department of Physics, College of Science and Humanities in Al-Kharj, Prince Sattam bin Abdulaziz University, Al-Kharj 11942, Saudi Arabia

⁶Department of Physics, Faculty of Science, Ain Shams University, Cairo, Egypt

⁷Department of Physics, Faculty of Science, Port Said University, Port Said 42521, Egypt

⁸Research Center for Physics (RCP), Department of Physics, Faculty of Science and Arts, Al-Mikhwah, Al-Baha University, Al-Baha 1988, Saudi Arabia

^{a)} Author to whom correspondence should be addressed: tantawy@sci.psu.edu.eg

ABSTRACT

The presence of the geometrical curvature makes the planar Korteweg–de Vries (KdV) equation inadequate to describe the propagation of nonlinear waves. In many scientific disciplines including plasma physics, nonlinear optics, oceanography, and communications, the cylindrical KdV (CKdV) equation becomes the appropriate choice for modeling these waves. Motivated by these applications, the Bäcklund transformation is used to analyze and find an analytical solution to the CKdV equation in the present investigation. For the first time, the multi-soliton solutions, including single-, two-, and three-soliton solutions, are investigated, and a general scheme is given to find N-soliton solutions of the CKdV equation in the context of plasma physics. Numerous researchers may find the given solutions helpful in understanding the mechanism of the generation of multi-solitons in their laboratory experiments and may also engender interest in the space physics community to look for these structures in the data coming from a variety of satellites roaming in space.

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I. INTRODUCTION

Several nonlinear integrable and non-integrable evolution equations (EEs) can be used for describing a wide variety of nonlinear phenomena (including solitons) in fluid dynamics, plasma physics, nonlinear optics, and many others.^{1–3} One of the primary normal modes of plasmas is the ion-acoustic (IA) waves (IAWs), which are driven by the inertia of the ions, whereas the electrons provide the restoring force and are considered massless.^{4,5} The propagation characteristics of this primary mode could alter in the presence of cold and hot electron populations. In such a case, an effective temperature ought

to be considered to describe the IA speed. The linear dispersion characteristics of the IAWs in the presence of two-electron temperatures (TETs) were studied by Jones *et al.*⁶ for the first time under the assumption that the timescale of the physical process responsible for causing the two thermal populations of electrons is much shorter than the ion timescale and, therefore, treated the two-electron components as two fluids. Since then, many investigations have been made, both theoretically^{6–21} and experimentally²² in this direction.

There have been many instances in geophysical and laboratory plasmas in the past few decades where the electrons have been found

to exhibit two velocity distributions. Solar wind at 1 AU, beam plasma systems, hot cathode discharge plasmas, and turbulent plasmas are a few such examples.²³ The satellite observations of the terrestrial bow shock²⁴ and the low and high entropy regions of interplanetary shocks^{25,26} have indicated the presence of two electron temperature (TET) plasmas. Observations of TET plasmas also include planetary magnetospheres,²⁷ heliospheric termination shock,²⁸ and a host of other regions in space.²⁹ GEOTAIL,³⁰ POLAR,³¹ S3–3,³² and FAST³³ have also observed TET species in the magnetosphere and aurora zone. Studies have shown that the features of the IAWs propagation are considerably changed in the presence of TET plasmas.^{34–36} Lakhina *et al.*³⁷ demonstrated that the increase in temperature of the hot electrons augments the amplitudes of both IA solitons (IASs) and electron-acoustic (EA) solitons in a TET plasma. In a complex, unmagnetized TET plasma, nonlinear propagation of Langmuir and IASs were also explored³⁸ and it was noted that these solitons travel at superacoustic speeds.

There have been plenty of satellite observations of electron velocity distributions in various areas of space plasmas that show a significant deviation from the Maxwellian distribution (MD).^{39,40} The notable examples include interstellar medium and auroral zone plasma,^{41–43} solar wind, and magnetosphere⁴⁴ as well as the terrestrial magnetosheath.^{45–48} Vasyliunas was the first scientist who made use of kappa distribution (KD) to interpret the data of Earth's magnetosphere obtained from OGO satellites.⁴⁹ Ever since KD has been gainfully employed to understand satellite data in Saturn, solar wind,⁵⁰ Jupiter, Earth's magnetospheric plasma sheet,⁵¹ and a variety of nonlinear phenomena observed in these regions.^{56,52–55} KD follows power law distribution and recovers the MD for low energy electrons, whereas the high energy electrons deviate from the MD. Schippers *et al.*⁵⁶ examined the data from the Cassini mission, which orbits Saturn and covers (5.4–20) R_s . In the magnetosphere of Saturn, the existence of cold and hot electron components with KD has been observed.^{14,18}

Although most of the studies were done in the planar geometry,^{36,52,57–62} it may not be realistic for many practical situations of interest in space and laboratory plasmas owing to the fact that the waves observed there are certainly unbounded in one-dimension (1-D). Accordingly, it is advantageous to take into account a nonplanar geometry in such situations.^{63–69} Examples where the nonplanar geometry is anticipated to play a significant role include shock tube (cylindrical geometry), capsule implosion (spherical geometry), and the interior of the stars.

In 1880, A. V. Bäcklund introduced the Bäcklund transformation (referred to as BT) as a pivotal concept in the context of the three-dimensional surfaces.⁷⁰ The BT primarily involves a system of first-order partial differential equations (PDEs) that establish connections between two distinct functions. In this context, both functions are termed as BTs of each other. Leveraging a known trivial solution, the BT becomes a mechanism for systematically generating a sequence of solutions for a given partial differential equation. Notably, the BT converts one nonlinear PDE into another with the requirement that the solution of the secondary equation remains consistent with the primary equation. Consequently, the application of the BT emerges as a robust avenue for finding solutions of intricate nonlinear PDEs.

A notable example of a BT is^{71,72} the Miura transformation, which serves to convert the Korteweg–de Vries (KdV) equation into a modified KdV (mKdV) equation.⁷⁰ Another facet of BT is the auto-

Bäcklund transformation (A-BT), characterized by its preservation of the invariance of the PDE. This transformative approach also facilitates the conversion of a known solution of a PDE into an alternate secondary solution for the same equation. This secondary solution subsequently serves as a stepping stone for deriving further solutions in a sequential manner.⁷⁰ An intriguing facet of this realm of study is the absence of a universally recognized systematic methodology for generating BTs.⁷⁰ Consequently, this subject remains a vibrant domain of research and exploration, as scholars continue to explore novel avenues for enhancing our understanding of these transformations.

Several decades ago, Nakamura⁷³ introduced the BT as an instrumental tool for finding the solution of the cylindrical KdV (CKdV) equation. Note that Nakamura and colleagues found the single and multi-soliton solutions of the CKdV equation using the bilinear BT, whereas we have obtained the single, two, and three-soliton solutions of the CKdV equation using the normal BT. It should be noted that the procedure to find multi-soliton solutions in our case differs significantly from the one adopted by Nakamura and co-authors. Furthermore, we have established a comprehensive mathematical framework for deriving N-soliton solutions of the CKdV equation. Interestingly, in contrast to the planar KdV equation, the solutions obtained for the CKdV equation exhibit a distinctive character: They manifest themselves as Airy's functions, introducing a notable layer of intricacy.

The article is organized in the following manner: A detailed illustration of the fluid model for IA waves (IAWs) is presented in Sec. II. The reductive perturbation technique (RPT) is employed for deriving the CKdV equation in Sec. III. Both single and multi-soliton solutions of the CKdV equation are presented in Secs. IV and V. The theoretically obtained results are applied to Saturn's magnetosphere, and the findings are reported in Sec. VI. Finally, the gist of the present work is given in Sec. VII.

II. ILLUSTRATION OF THE FLUID MODEL

Employing the multi-fluid theory, we consider an unmagnetized collisionless plasma comprising non-Maxwellian two-temperature electrons and inertial ions. Additionally, it is supposed that nonlinear waves move only in the radial direction. The equilibrium condition for plasma is $n_{i0} = n_{c0} + n_{h0}$, where n_{i0} , n_{c0} , and n_{h0} signify the unperturbed concentrations of positive ions and cold and hot electrons, respectively. For investigating the dynamics of the IAWs in the current model, the following dimensionless fluid equations are considered:

$$\frac{\partial n_i}{\partial t} + \frac{1}{r} \frac{\partial}{\partial r} (r n_i v_i) = 0, \quad (2.1)$$

$$\frac{\partial v_i}{\partial t} + v_i \frac{\partial v_i}{\partial r} = -\frac{\partial \Phi}{\partial r}, \quad (2.2)$$

and

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \Phi}{\partial r} \right) = \mu n_c + \nu n_h - Z_i n_i. \quad (2.3)$$

The effective electron temperature is defined as $T_{eff} = T_{eh}/v \left(+\beta \frac{n_{h0}}{n_{i0}} \right)$, and the effective sound speed is given as $c_{s,eff} = (Z_i T_{eff}/m_i)^{1/2}$. The space and time scales are normalized with effective Debye length $\lambda_{D,eff} = (T_{eff}/4\pi Z_i n_{i0} e^2)^{1/2} \equiv c_{s,eff}/\omega_{pi}$ and inverse ion plasma

frequency $\omega_{pi}^{-1} = (4\pi n_{i0} Z_i^2 e^2 / m_i)^{-1/2}$, respectively.³⁶ Also, the other quantities are normalized as $v = v_i / c_{s,eff}$, $n_i = n_i / n_{i0}$, $\Phi = \phi / T_{eff} / e$. Here, $v = n_{h0} / n_{i0}$, $\mu = n_{c0} / n_{i0}$, $\beta = T_c / T_h$, and $Z_i = 1$.

We get the following normalized forms of the total number densities of both cold and hot electrons by integrating over velocity space for the kappa κ distribution (see the Appendix):

$$n_c = 1 + A\delta\Phi + B\delta^2\Phi^2, \quad (2.4)$$

$$n_h = 1 + A_1\alpha\Phi + B_1\alpha^2\Phi^2, \quad (2.5)$$

with

$$A_1 = \frac{b_1}{a_1} \quad \text{and} \quad B_1 = \frac{b_1 c_1}{2a_1^2}, \quad (2.6a)$$

$$A = \frac{b}{a} \quad \text{and} \quad B = \frac{bc}{2a^2}, \quad (2.6b)$$

where spectral indices of the hot and cold electrons are represented by κ_h and κ_c respectively. $a_1 = (\kappa_h - 3/2)$, $b_1 = (\kappa_h - 1/2)$, $c_1 = (\kappa_h + 1/2)$, $a = (\kappa_c - 3/2)$, $b = (\kappa_c - 1/2)$, $c = (\kappa_c + 1/2)$, and for $(\kappa_h, \kappa_c) \rightarrow \infty$, the KD reduces to the well-known MD, which leads to $(A_1, B_1, C_1) = (A, B, C) = (1, 1/2, 1/6)$. The temperature ratios are written as $\alpha = T_{eff} / T_h$ and $\delta = T_{eff} / T_c$.

III. DERIVATION OF ONE-DIMENSIONAL NONLINEAR DYNAMICAL EQUATION (CKDV EQUATION)

Here, the small but finite amplitude IA solitary waves (IASWs) moving in the radial direction are examined using the weak turbulence theory. Correspondingly, the following stretched coordinates for space and time variables are considered:

$$\xi = \epsilon^{1/3}(r - \lambda t) \quad \text{and} \quad \tau = \epsilon^{2/3}t, \quad (3.1)$$

where λ is the phase velocity of the IAW normalized with $c_{s,eff}$, and ϵ is a small and real parameter ($0 < \epsilon < 1$), which indicates the strength of the nonlinearity. The physical quantities n_i , v_i , and Φ are expanded as

$$n_i = 1 + \epsilon n_{i1} + \epsilon^2 n_{i2} + \epsilon^3 n_{i3} + \dots, \quad (3.2a)$$

$$v_i = \epsilon v_{i1} + \epsilon^2 v_{i2} + \epsilon^3 v_{i3} + \dots, \quad (3.2b)$$

$$\Phi = \epsilon\Phi^{(1)} + \epsilon^2\Phi^{(2)} + \epsilon^3\Phi^{(3)} + \dots \quad (3.2c)$$

Using Eqs. (2.4), (2.5), (3.1), and (3.2a)–(3.2c) in Eqs. (2.1)–(2.3) and collecting the terms of lowest order in ϵ , we obtain

$$v_{i1} = \frac{\Phi^{(1)}}{\lambda}, \quad (3.3a)$$

$$n_{i1} = \frac{\Phi^{(1)}}{\lambda^2}, \quad (3.3b)$$

$$n_{i1} = (\mu\delta A + \nu\alpha A_1)\Phi^{(1)}. \quad (3.3c)$$

Solving Eqs. (3.3a)–(3.3c) simultaneously yields the following relation:

$$\lambda = \sqrt{\frac{1}{(\mu\delta A + \nu\alpha A_1)}}. \quad (3.4)$$

Equation (3.4) represents the linear dispersion relation of the IAWs of the current plasma model. The impact of the plasma parameters appears through the coefficients A and A_1 .

Collecting the order of $\epsilon^{2/3}$, we arrive at the following set of equations:

$$-\lambda \frac{\partial n_{i2}}{\partial \xi} + \frac{\partial v_{i2}}{\partial \xi} + \frac{v_{i1}}{\lambda \tau} + \frac{\partial n_{i1}}{\partial \tau} = -\frac{\partial(n_{i1} v_{i1})}{\partial \xi}, \quad (3.5a)$$

$$-\lambda \frac{\partial v_{i2}}{\partial \xi} + \frac{\partial v_{i1}^2}{\partial \xi} + \frac{\partial v_{i1}}{\partial \tau} = -\frac{\partial \Phi^{(2)}}{\partial \xi}, \quad (3.5b)$$

$$n_{i2} - (\mu\delta A + \nu\alpha A_1)\Phi^{(2)} = -\frac{\partial^2 \Phi^{(1)}}{\partial \xi^2} + (\mu\delta^2 B + \alpha^2 \nu B_1)\Phi_1^{(2)}. \quad (3.5c)$$

Solving Eqs. (3.5a)–(3.5c) simultaneously yields the following CKdV equation:

$$\frac{\partial \Phi^{(1)}}{\partial \tau} + C\Phi^{(1)} \frac{\partial \Phi^{(1)}}{\partial \xi} + D \frac{\partial^3 \Phi^{(1)}}{\partial \xi^3} + \frac{\Phi^{(1)}}{2\tau} = 0. \quad (3.6)$$

Here, the coefficients C and D represent nonlinearity and dispersion, respectively, and are given by

$$C = \frac{3}{2\lambda} - \lambda^3 (\mu\delta^2 B + \alpha^2 \nu B_1), \quad (3.7a)$$

$$D = \frac{\lambda^3}{2}, \quad (3.7b)$$

whereas the final term represents the contribution of the cylindrical geometry. By making the transformation $\Phi^{(1)} = 6C^{-1}D^{1/3}u$, $\xi = D^{1/3}x$, and $\tau = t$ to the CKdV [Eq. (3.6)], we obtain the following standard CKdV equation:

$$u_t + 6uu_x + u_{xxx} + \frac{u}{2t} = 0. \quad (3.8)$$

In the study of solitons in a one-dimensional system, the simplest and thus important model is provided by the CKdV equation, which was first studied by Maxon and Viecelli.^{73–75} The CKdV equation admits infinite conservation laws.⁷⁶ In Eq. (3.8), lower suffixes, x and t , represent partial differentiation.

IV. BÄCKLUND TRANSFORMATION (BT) OF THE CKDV EQUATION

We define two solutions w and w' of the CKdV equation by $u = -w_x$, $u' = -w'_x$ or $w = -(2 \log f)_x$, $w' = -(2 \log f')_x$.⁷³ Here, prime notation does not represent the derivative.

So, the BT for the CKdV equation can be written as follows:⁷³

$$(W'_+)_x = \frac{1}{2}(-W'_-)^2 - \frac{x+x_1}{6t}, \quad (4.1a)$$

$$(-W'_-)_t = (W'_-)_{xxx} - \frac{3}{2}\{(W'_-)(W'_+)_x\}_x + \frac{1}{4}\{(W'_-)^3\}_x + \left\{\frac{x+x_1}{4t}(W'_-)\right\}_x, \quad (4.1b)$$

where $W'_- \equiv (w-w')$ and $W'_+ \equiv (w+w')$.

Here, the constant x_1 , which is arbitrary chosen, has a physical interpretation as the initial position of the soliton.⁷³ This can be easily checked that the set of Eqs. (4.1a) and (4.1b) satisfies the integrability condition that is [i.e., $(W'_+)_{xt} = (-W'_-)_{tx}$]. Starting from vacuum solution $w' = 0$ and $x_1 = 0$ in Eqs. (4.1a) and (4.1b), we obtain the following equations:

$$\frac{dw}{dx} = \frac{1}{2}w^2 - \frac{x}{6t}, \quad (4.2)$$

$$\begin{aligned} \frac{dw}{dt} = & -\frac{d^3w}{dx^3} + \frac{3}{2}\left(\frac{dw}{dx}\right)^2 + \frac{3}{2}w\left(\frac{d^2w}{dx^2}\right) \\ & - \frac{3}{4}w^2\left(\frac{dw}{dx}\right) - \frac{x}{4t}\left(\frac{dw}{dx}\right) - \frac{w}{4t}. \end{aligned} \quad (4.3)$$

By using Eqs. (4.2) and (4.3), similarity transformation pair can be written as follows:

$$\eta = \frac{x}{t^{1/3}}, \quad (4.4a)$$

$$w = t^{-1/3}g(\eta). \quad (4.4b)$$

Equations (4.2) and (4.3) in the new variables η and $g(\eta)$, respectively, can be written as follows:

$$\frac{dg}{d\eta} = \frac{1}{2}g^2 - \frac{\eta}{6}, \quad (4.5)$$

$$\frac{d^3g}{d\eta^3} - \frac{3}{2}\left(\frac{dg}{d\eta}\right)^2 - \frac{3}{2}g\frac{d^2g}{d\eta^2} = \eta\frac{dg}{d\eta} - \frac{3}{4}g^2\frac{dg}{d\eta} - \frac{g}{4}. \quad (4.6)$$

Now, Eq. (4.5) is quite similar to the “Riccati equation.” The general Riccati equation is given by

$$\frac{dg}{d\eta} = a(\eta)g^2 + b(\eta)g + c(\eta). \quad (4.7)$$

Now, we will do the appropriate substitution

$$g(\eta) = -\frac{1}{a(\eta)T}\frac{dT}{d\eta}, \quad (4.8)$$

and from Eqs. (4.5)–(4.8), we can easily write the following equation:

$$\frac{d^2T}{d\eta^2} - \frac{1}{12}\eta T = 0. \quad (4.9)$$

Equation (4.9) is the standard “Airy’s equation.” The solution of Eq. (4.9) reads as

$$T(\eta) = c_1 Ai\left(\frac{\eta}{2^{2/3} \cdot 3^{1/3}}\right) + c_2 Bi\left(\frac{\eta}{2^{2/3} \cdot 3^{1/3}}\right), \quad (4.10)$$

where c_1 and c_2 are arbitrary constants. For simplicity, choose $c_1 = c_2 = 1$. Now, the value of $g(\eta)$ can be easily found from Eq. (4.8), and by using Eqs. (4.4a) and (4.4b), the final solution w of Eq. (4.2) comes out to be

$$w(x, t) = -t^{1/3} \frac{\left(\frac{1}{2^{2/3} \cdot 3^{1/3}}\right) \left[Ai'\left(\frac{x/t^{1/3}}{2^{2/3} \cdot 3^{1/3}}\right) + Bi'\left(\frac{x/t^{1/3}}{2^{2/3} \cdot 3^{1/3}}\right) \right]}{\frac{1}{2} \left[Ai\left(\frac{x/t^{1/3}}{2^{2/3} \cdot 3^{1/3}}\right) + Bi\left(\frac{x/t^{1/3}}{2^{2/3} \cdot 3^{1/3}}\right) \right]}. \quad (4.11)$$

The above-mentioned solution in terms of x_1 can be written as

$$w_1(x, t) = -t^{1/3} \frac{\left(\frac{1}{2^{2/3} \cdot 3^{1/3}}\right) \left[Ai'\left(\frac{(x+x_1)/t^{1/3}}{2^{2/3} \cdot 3^{1/3}}\right) + Bi'\left(\frac{(x+x_1)/t^{1/3}}{2^{2/3} \cdot 3^{1/3}}\right) \right]}{\frac{1}{2} \left[Ai\left(\frac{(x+x_1)/t^{1/3}}{2^{2/3} \cdot 3^{1/3}}\right) + Bi\left(\frac{(x+x_1)/t^{1/3}}{2^{2/3} \cdot 3^{1/3}}\right) \right]}. \quad (4.12)$$

According to the transformation $u = -w_{1,x}$ the analytical solution of the CKdV equation reads as

$$\begin{aligned} u(x, t) = & \left[\frac{\left(\frac{2}{3}\right)^{1/3} \left(\frac{1}{t}\right)^{1/3} \left(\frac{\left(\frac{1}{t}\right)^{2/3} (x+x_1) Ai\left[\frac{(x+x_1)/t^{1/3}}{2^{2/3} \cdot 3^{1/3}}\right] + \left(\frac{1}{t}\right)^{2/3} (x+x_1) Bi\left[\frac{(x+x_1)/t^{1/3}}{2^{2/3} \cdot 3^{1/3}}\right]}{2^{4/3} \cdot 3^{2/3}} \right)}{Ai\left(\frac{(x+x_1)/t^{1/3}}{2^{2/3} \cdot 3^{1/3}}\right) + Bi\left(\frac{(x+x_1)/t^{1/3}}{2^{2/3} \cdot 3^{1/3}}\right)} \right] \\ & - \left[\frac{\left(\frac{2}{3}\right)^{1/3} \left(\frac{1}{t}\right)^{1/3} \left(Ai'\left[\frac{(x+x_1)/t^{1/3}}{2^{2/3} \cdot 3^{1/3}}\right] + Bi'\left[\frac{(x+x_1)/t^{1/3}}{2^{2/3} \cdot 3^{1/3}}\right] \right) \left(\left(\frac{1}{t}\right)^{1/3} Ai'\left[\frac{(x+x_1)/t^{1/3}}{2^{2/3} \cdot 3^{1/3}}\right] + \left(\frac{1}{t}\right)^{1/3} Bi'\left[\frac{(x+x_1)/t^{1/3}}{2^{2/3} \cdot 3^{1/3}}\right] \right)}{\left(Ai\left[\frac{(x+x_1)/t^{1/3}}{2^{2/3} \cdot 3^{1/3}}\right] + Bi\left[\frac{(x+x_1)/t^{1/3}}{2^{2/3} \cdot 3^{1/3}}\right] \right)^2} \right]. \end{aligned} \quad (4.13)$$

Equation (4.13) is the one-soliton solution of the CKdV equation [i.e., Eq. (3.8)].

V. THEOREM OF PERMUTABILITY (TP)

The permutability theorem, as stated in Ref. 77 asserts that two consecutive Bäcklund transformations are commutative. This can be

expressed more precisely as follows: If two BTs with different parameters, x_1 and x_2 , are applied to a solution w , resulting in a new solution w_{12} , then the order in which the transformations are applied does not matter. In other words, the same final solution is obtained regardless of whether the first or second transformation is applied first.⁷⁷ Namely, if

$$w \xrightarrow{x_1} w_1 \xrightarrow{x_2} w_{12} \quad (5.1)$$

and

$$w \xrightarrow{x_2} w_2 \xrightarrow{x_1} w_{21}, \quad (5.2)$$

then, of course, the TP requires

$$w_{12} = w_{21}. \quad (5.3)$$

Quantitatively, from the BT in Eq. (4.1a) and after applying the TP, we obtain the following two-soliton solution:

$$w_{12} = w + \frac{x_1 - x_2}{3t(w_1 - w_2)}. \quad (5.4)$$

Note that like w_1 [see Eq. (4.12)], the expression of w_2 can be written as follows:

$$w_2(x, t) = -t^{1/3} \frac{\left(\frac{1}{2^{2/3} \cdot 3^{1/3}} \right) \left[Ai' \left(\frac{(x+x_2)/t^{1/3}}{2^{2/3} \cdot 3^{1/3}} \right) + Bi' \left(\frac{(x+x_2)/t^{1/3}}{2^{2/3} \cdot 3^{1/3}} \right) \right]}{\frac{1}{2} \left[Ai \left(\frac{(x+x_2)/t^{1/3}}{2^{2/3} \cdot 3^{1/3}} \right) + Bi \left(\frac{(x+x_2)/t^{1/3}}{2^{2/3} \cdot 3^{1/3}} \right) \right]}. \quad (5.5)$$

It is important to observe that Eq. (5.4) maintains symmetry when indices 1 and 2 are interchanged, in accordance with the requirement of TP, and that

$$u_{12}(x, t) = -\frac{\partial w_{12}(x, t)}{\partial x}. \quad (5.6)$$

The two-soliton solution of Eq. (3.8) can be obtained by using Eq. (5.6) and by letting $w = 0$ in Eq. (5.4) and reads as follows:

$$u_{12}(x, t) = \frac{(x_1 - x_2) \left[-\frac{C \left(\frac{D \cdot A_1 + D \cdot B_1}{2^{4/3} \cdot 3^{2/3}} \right)}{A_1 + B_1} + \frac{C \left(\frac{E \cdot A_2 + E \cdot B_2}{2^{4/3} \cdot 3^{2/3}} \right)}{A_2 + B_2} + \frac{C(A'_1 + B'_1) \left(\frac{(\frac{1}{t})^{1/3} A'_1 + (\frac{1}{t})^{1/3} B'_1}{2^{2/3} \cdot 3^{1/3}} \right)}{(A_1 + B_1)^2} - \frac{C(A'_2 + B'_2) \left(\frac{(\frac{1}{t})^{1/3} A'_2 + (\frac{1}{t})^{1/3} B'_2}{2^{2/3} \cdot 3^{1/3}} \right)}{(A_2 + B_2)^2} \right]}{3t \left(-\frac{C(A'_1 + B'_1)}{A_1 + B_1} + \frac{C(A'_2 + B'_2)}{A_2 + B_2} \right)^2}, \quad (5.7)$$

with

$$\begin{aligned} A_1 &= Ai \left(\frac{\left(\frac{1}{t} \right)^{1/3} (x+x_1)}{2^{2/3} \cdot 3^{1/3}} \right), & B_1 &= Bi \left(\frac{\left(\frac{1}{t} \right)^{1/3} (x+x_1)}{2^{2/3} \cdot 3^{1/3}} \right), & A_2 &= Ai \left(\frac{\left(\frac{1}{t} \right)^{1/3} (x+x_2)}{2^{2/3} \cdot 3^{1/3}} \right), & B_2 &= Bi \left(\frac{\left(\frac{1}{t} \right)^{1/3} (x+x_2)}{2^{2/3} \cdot 3^{1/3}} \right), \\ A'_1 &= Ai' \left(\frac{\left(\frac{1}{t} \right)^{1/3} (x+x_1)}{2^{2/3} \cdot 3^{1/3}} \right), & B'_1 &= Bi' \left(\frac{\left(\frac{1}{t} \right)^{1/3} (x+x_1)}{2^{2/3} \cdot 3^{1/3}} \right), & A'_2 &= Ai' \left(\frac{\left(\frac{1}{t} \right)^{1/3} (x+x_2)}{2^{2/3} \cdot 3^{1/3}} \right), & B'_2 &= Bi' \left(\frac{\left(\frac{1}{t} \right)^{1/3} (x+x_2)}{2^{2/3} \cdot 3^{1/3}} \right), \\ C &= \left(\frac{2}{3} \right)^{1/3} \left(\frac{1}{t} \right)^{1/3}, & D &= \left(\frac{1}{t} \right)^{2/3} (x+x_1), & E &= \left(\frac{1}{t} \right)^{2/3} (x+x_2). \end{aligned}$$

Figure 1 gives the schematic diagram for constructing the two-soliton solution. The permutability theorem enables us to construct higher-order solutions algebraically at every given order.⁷⁷

Following the aforementioned procedure, the general expression for the solution at the n th step reads as

$$w_{(n)} = w_{(n-2)} + \frac{x_{(n-1)} - x_{(n)}}{3t(w_{(n-1)} - w'_{(n-1)})}, \quad (5.8)$$

$$w_{(n)} = w_{\{k_1, k_2, \dots, k_n\}} = w_{12 \dots n}, \quad (5.9)$$

$$w'_{(n)} = w_{\{k_1, \dots, k_{n-1}, k_{n+1}\}} = w_{12 \dots n-1, n+1}, \quad (5.10)$$

and

$$w = w_{(0)} = 0. \quad (5.11)$$

Similarly, the three-soliton solution can be written as

$$w_{123} = w_1 + \frac{x_2 - x_3}{3t(w_{12} - w_{13})}, \quad (5.12)$$

where w_1 is the first-soliton solution. The schematics for constructing the three-soliton solution are given in Fig. 2.

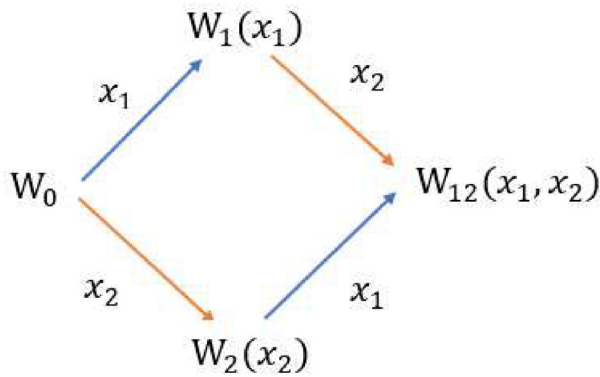


FIG. 1. Bianchi diagram for constructing the two-soliton solution. We start from a vacuum solution $w_0 = 0$.

VI. RESULTS AND DISCUSSION

Here, we present the results of the nonlinear propagation of IAWs in a cylindrical geometry in the presence of hot and cold electrons that are considered to be KD in view of Cassini's observations of Saturn's magnetosphere.⁵⁶ For the physical model under consideration, the CKdV equation admits compressive IA solitary structures. Unlike the normal KdV equation, the solution of the CKdV equation is in the form of Airy's functions of first and second kind, i.e., Ai and Bi as shown in Eq. (5.12). We plot here the single soliton solution by varying the important plasma parameters of the current model. Note that we have also given the mathematical expressions for two and three-soliton solutions of the CKdV equation. Furthermore, we have given a recipe to construct N-soliton solutions of the CKdV equation. We would like to mention here that plotting the multi-soliton solutions is an extremely arduous task owing to the presence of Airy's functions of first and second kind; however, it is a task worth pursuing in the future.

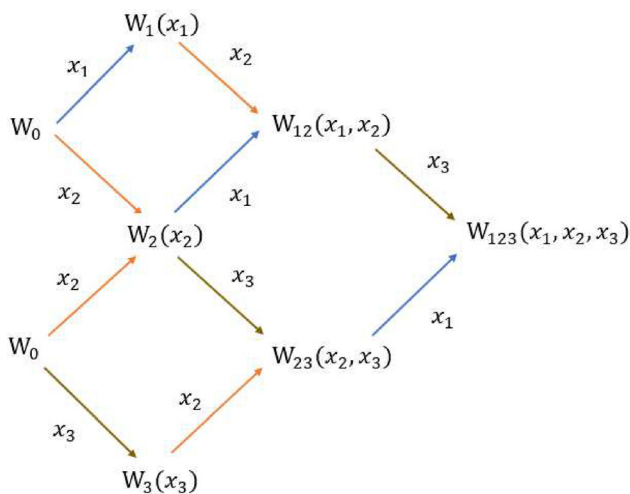


FIG. 2. Bianchi diagram for constructing the three-soliton solution. We start from a vacuum solution, and by following various paths, we reach the three-soliton solution w_{123} .

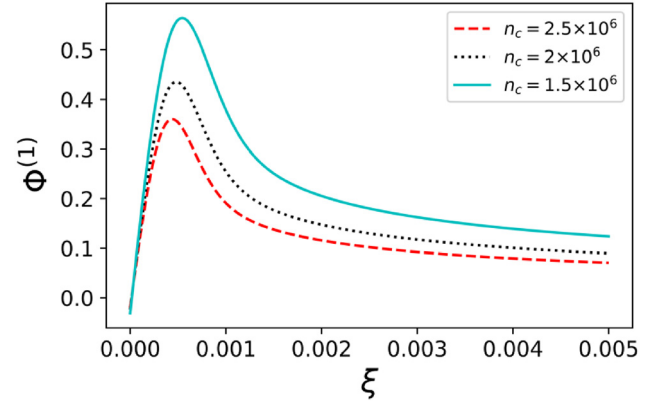


FIG. 3. Variation of the CKdV soliton for different values of cold electron density.

A. Effect of cold electron density

Figure 3 manifests the behavior of a single cylindrical IA soliton (CIAS) by varying the cold electron number density n_c . It is noted that the amplitude of the CIAS decreases by increasing the cold electron number density, which happens due to the increasing percentage of the cold electrons which results in a decrease in the temperature of the plasma. We have illustrated these findings for specific values, namely, $n_c = 2.5 \text{ m}^{-3}$, $n_c = 2 \text{ m}^{-3}$, and $n_c = 1.5 \text{ m}^{-3}$.¹⁵

B. Effect of hot electron density n_h

Figure 4 explores the impact of hot electron number density n_h on the behavior of single CIASs. The enhancing hot electron number density is found to augment the amplitude of the CIASs owing to the fact that the increasing percentage of hot electrons enhances the temperature of the plasma. We have illustrated these findings for specific values, namely, $n_h = 0.18 \text{ m}^{-3}$, $n_h = 0.11 \text{ m}^{-3}$, and $n_h = 0.07 \text{ m}^{-3}$.¹⁵

C. Effect of κ distribution

Figure 5 shows a detailed comparison of the amplitude of CIASs for kappa and Maxwellian distributions for cold and hot electrons.

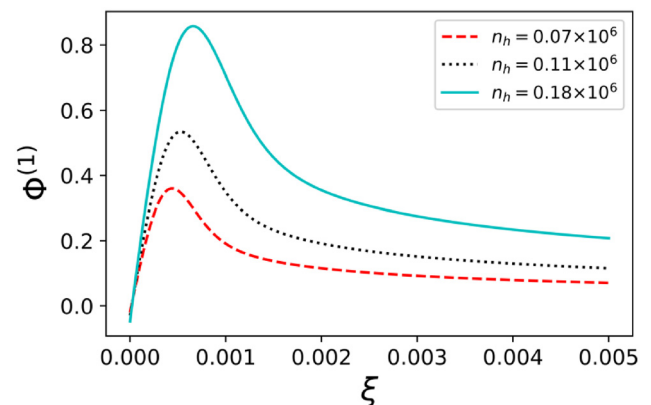


FIG. 4. Variation of the CKdV soliton for different values of hot electron density.

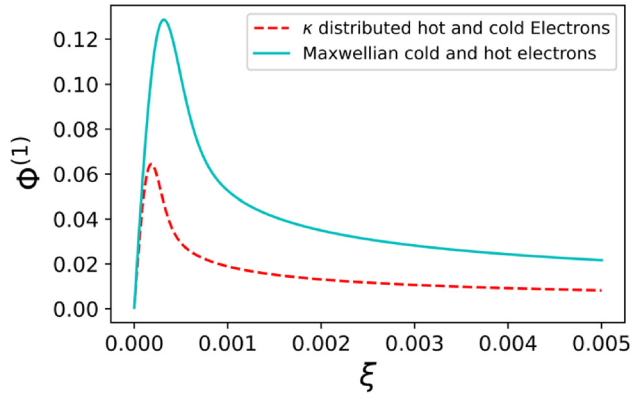


FIG. 5. Comparison between Maxwellian and kappa distributions for the CKdV soliton.

It must be noted that cold and hot kappa electrons approach the Maxwellian distribution in the limit $(\kappa_c, \kappa_h) \rightarrow \infty$. The amplitude of the CIAS for the Maxwellian cold and hot electrons is found to be much higher than its kappa counterpart, which indicates that the Maxwellian distribution energizes the CIASs. It can be seen from Fig. 5 that the width of the CIAS for kappa cold and hot electrons is much smaller than the Maxwellian ones, which essentially implies that the CIASs form over a very short spatial scale for the kappa electrons as opposed to the Maxwellian case. We have illustrated these findings for specific values, namely, for non-Maxwellian ($\kappa_c = 2$ and $\kappa_h = 3$)¹⁵ and for Maxwellian (A_1, B_1, C_1, A, B , and C) = (1, 1/2, 1/6).

D. Effect of electron spectral indices κ_c and κ_h

Figure 6 shows the behavior of a single cylindrical IA soliton (CIAS) for increasing values of κ_c . It is found that enhancing κ_c enhances the amplitude of the CIAS. This is due to the fact that the increase in κ_c is tantamount to increasing the temperature of the cold electrons or energization of the system. We have illustrated these findings for specific values, namely, $\kappa_c = 2$, $\kappa_c = 2.1$, and $\kappa_c = 2.3$.¹⁵ In Fig. 7, note that the increasing κ_h does not alter the structure of the CIASs as the number density of hot electrons is much smaller than the cold

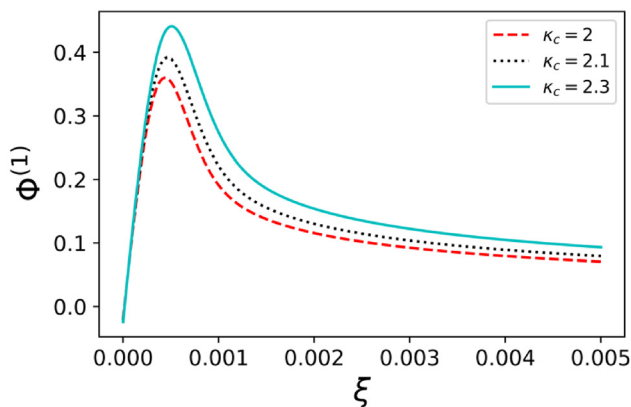


FIG. 6. Variation of the CKdV soliton for different values of κ_c .

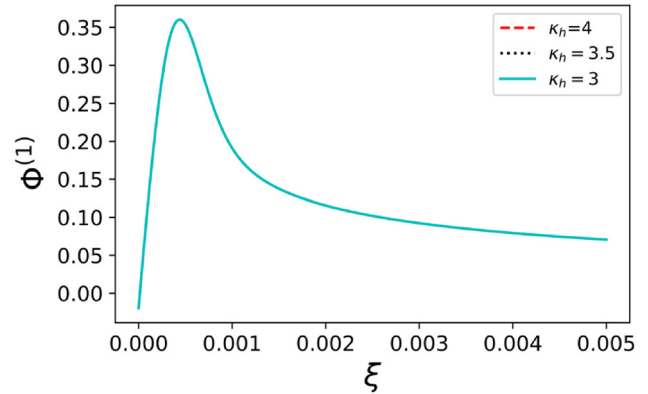


FIG. 7. Variation of the CKdV soliton for different values of κ_h .

electrons, and hence, the effect is negligible. We have illustrated these findings for specific values, namely, $\kappa_h = 4$, $\kappa_h = 3.5$, and $\kappa_h = 3$,¹⁵ as shown in Fig. 7.

E. Effect of different sets of plasma parameters at different Saturn radii

Figure 8 manifests the effect of different sets of values of plasma parameters at different Saturn radii (see the table given in Ref. 15) on the propagation characteristics of CIASs. It is seen that the change of plasma parameters at different Saturn radii of R_s (where $R_s = 60\,300$ km is the radius of Saturn) leads to the enhancement of the amplitude and width of the CIASs owing to the presence of different sets of plasma parameters at different Saturn radii, and hence, the spatial scales over which CIASs form vary accordingly. We have illustrated these findings for specific values, namely, $R_s = 5.4$, $R_s = 9.8$, and $R_s = 12$.¹⁵

VII. CONCLUSION

In this paper, one-dimensional propagation of nonlinear electrostatic waves has been investigated in a cylindrical geometry on the ion timescale in the presence of a two-electron population in a collisionless and unmagnetized plasma. Cylindrical KdV (CKdV) has been derived

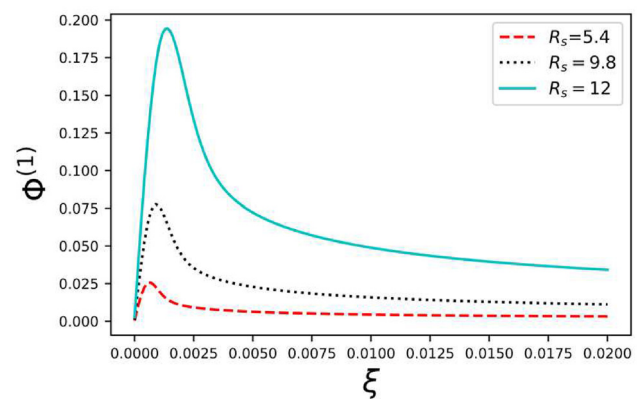


FIG. 8. Variation of the CKdV soliton for different values of R_s .

for the system under investigation using the RPT. Employing the BT, single, two, and three-soliton solutions have been presented, and a general expression has been given to find N -soliton solutions of the CKdV equation. The single soliton solution of the CKdV equation has been expressed in terms of Airy's functions and has been applied in plasma physics for the first time to the best of our knowledge. It has been shown that the percentage of hot and cold electrons and the values of κ_c and κ_h affect the structure of the cylindrical ion-acoustic solitons (CIASs) appreciably. It has also been shown that at different Saturn radii, different sets of plasma parameters are present and, therefore, the propagation characteristics and the spatial scales over which the CIASs form vary throughout Saturn's magnetosphere. The challenges concerning the graphical representation of the interaction of CIASs have also been reported.

ACKNOWLEDGMENTS

The authors express their gratitude to Princess Nourah bint Abdulrahman University Researchers supporting Project No. PNURSP2023R32, Princess Nourah bint Abdulrahman University, Riyadh, Saudi Arabia. This study was supported via funding from Prince Sattam bin Abdulaziz University Project No. PSAU/2023/R/1444. One of the authors, W. Masood, would like to acknowledge support from the ICTP through the Associates Programme (2018–2023).

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

All authors contributed equally and approved the final manuscript.

M. Shan Tariq: Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Writing – original draft (equal). **Waqas Masood:** Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Software (equal); Supervision (equal); Writing – original draft (equal). **M. Siddiq:** Conceptualization (equal); Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal). **S. Asghar:** Formal analysis (equal); Investigation (equal); Methodology (equal); Software (equal). **B. M. Alotaibi:** Conceptualization (equal); Data curation (equal); Formal analysis (equal). **Sherif M. E. Ismaeel:** Formal analysis (equal); Investigation (equal); Writing – review & editing (equal). **S. A. El-Tantawy:** Formal analysis (equal); Investigation (equal); Supervision (equal); Validation (equal); Writing – review & editing (equal).

DATA AVAILABILITY

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

APPENDIX: SOME DETAILS FOR SUPERHERMAL ELECTRON DENSITIES

For kappa distribution, the following generalized three-dimensional distribution function $f_\kappa(v)$ is given by^{78–80}

$$f_\kappa(v) = A_\kappa \left(1 + \frac{v^2}{\kappa \theta^2} \right)^{-(1+\kappa)}, \quad (\text{A1})$$

with

$$A_j^{(\kappa)} = \frac{n^{(0)}}{\sqrt{(\pi \kappa \theta^2)^3} \Gamma\left(\kappa + \frac{1}{2}\right)}, \quad (\text{A2})$$

$$\theta = \sqrt{\left[\frac{\kappa - \frac{3}{2}}{\kappa} \right] v_{th}^2}, \quad \text{and} \quad v_{th} = \sqrt{\frac{2K_B T}{m}},$$

where $v^2 = v_x^2 + v_y^2 + v_z^2$ is the velocity of the moving species, A_κ indicates the constant of normalization, θ denotes an effective thermal speed, and v_{th} represents the thermal speed, whereas m and κ are the mass of species under consideration and the spectral index, respectively. Physically, the value of the spectral index κ must be larger than $3/2$. Let us take a quick physical look at this distribution. For $v \gg 0$, the current distribution has a high-velocity tail, which decreases with v as a power law: $f_\kappa(v) \propto v^{-2(1+\kappa)}$. On the contrary, for $v < \theta$ or comparable to θ , $f_\kappa(v)$ is close to the Maxwellian velocity distribution function. Now, by integrating the kappa distribution function over velocity space $n_{e,j} \equiv n_j = \int f_\kappa(v) d^3v$, the following number density of the electrons is obtained:

$$n_j = n_j^{(0)} \left[1 - \frac{e\phi}{(\kappa_j - \frac{3}{2}) K_B T_j} \right]^{-\kappa_j + \frac{1}{2}}. \quad (\text{A3})$$

Note here that $j=c$ and $j=h$ for cold and hot electrons, respectively.

In the normalized form, the electron number densities read as follows:

$$n_j = \left[1 - \frac{\sigma_j \Phi}{a_j} \right]^{-b_j}, \quad (\text{A4})$$

where $a_j = (\kappa_j - \frac{3}{2})$, $b_j = \kappa_j - \frac{1}{2}$, and $\sigma_j = T_{eff}/T_j$, where T_{eff} indicates the effective temperature of the electrons.

Expanding n_j given in Eq. (A4) around its equilibrium value, we get

$$n_j = 1 + S_1(\sigma_j \Phi) + S_2(\sigma_j \Phi)^2 + \dots, \quad (\text{A5})$$

with

$$S_1 = \frac{b_j}{a_j} \quad \text{and} \quad S_2 = \frac{b_j(1+b_j)}{2a_j^2}.$$

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